

ASAD HUSSAIN

SOFTWARE ENGINEER | FULL STACK DEVELOPER | AI AUTOMATION

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GitHub: github.com/asadh-74 | Automation Portfolio



PROFESSIONAL SUMMARY

Software Engineer and Full Stack Developer with experience across four internships in backend engineering, AI and automation. Builds web applications, database-backed APIs and workflows that connect live information to practical business tools. Combines Python, JavaScript and SQL with modern agent frameworks to develop bid-monitoring platforms, e-commerce experiences and intelligent applications. Brings a delivery-focused approach to API integration, testing, deployment and clear technical documentation.

AREAS OF EXPERTISE

Full Stack Development: JavaScript, HTML5, CSS3, React, Vue.js, Python, FastAPI, Flask; responsive interfaces, REST APIs and frontend/backend integration.

Backend & Data: SQL, SQL Server, MySQL, SQLite, database modeling, CRUD services, authentication, validation; PHP/Laravel API fundamentals.

AI & Automation: LangChain, LangGraph, CrewAI, MCP servers, n8n, Zapier, Playwright, Firecrawl, OpenClaw; LLM integration, RAG and prompt engineering.

Engineering Tools: Git/GitHub, Docker, Linux, Postman, CI/CD, GCP; TensorFlow, PyTorch, Scikit-learn, OpenCV, Streamlit and ESP32 integration.

EDUCATION

National University of Sciences and Technology (NUST)

Islamabad Campus | B.E. Electrical Engineering | 2023 - 2027 (expected)

INTERNSHIP EXPERIENCE

Full Stack Development Intern | Lantrotech

Gulberg Greens, Islamabad | July 2026 - August 2026

- Developed a full stack bid-monitoring project connecting a frontend dashboard, backend services and SQL storage for live source data and procurement opportunities.
- Used Playwright and Firecrawl in data-collection workflows, with automation and MCP tool integrations supporting collection, processing and access to structured records.
- Outcome: brought bid information into a centralized application for searching, monitoring and reviewing opportunities with less repeated manual browsing.

Application Development Intern | Codes Apex (Pvt.) Ltd.

Islamabad | June 2025 - August 2025

- Contributed to application development across user interfaces, backend functionality and data-driven workflows, translating requirements into practical application features.
- Worked on integration, debugging and functional checks; gained experience connecting application screens with business logic and persistent data.

Backend AI Engineering Intern | FlyRank AI

Chicago, USA (Remote) | 1 July 2026 - 7 September 2026

- Built an authentication-protected CRUD REST API backed by a relational database and containerized the application with Docker for reproducible deployment.
- Automated a data-scraping workflow and applied prompt engineering and LLM tooling to search-intelligence tasks, including quality scoring and review ranking.

Machine Learning Intern | CodeAlpha

Remote | 20 June 2026 - 20 July 2026

- Developed credit-scoring, handwritten-character recognition and disease-prediction models using classical classifiers and convolutional neural networks.
- Compared model families using precision, recall, F1-score and ROC-AUC; explored speech-emotion recognition with MFCC features and MLP/CNN models.

SELECTED SOFTWARE & AI PROJECTS

Bid Monitoring & Tender Intelligence Platform

Python | Playwright | Firecrawl | SQL | MCP servers | Full-stack dashboard

- Built during the Lantrotech internship to collect bid information from web sources and connect automated ingestion with database-backed application views.
- Structured the workflow around source extraction, data processing and persistent SQL records, making updated opportunities available for dashboard review.
- Outcome: a consolidated workflow for opportunity discovery and monitoring, supporting quicker access to relevant bid information.

Aeterna Watches - E-Commerce Platform

HTML/CSS | JavaScript | REST APIs | LLM integration | Vercel

- Developed a responsive watch storefront with a product catalog and checkout flow, connecting the shopping interface with backend services.
- Integrated an AI shopping concierge using Gemini/OpenAI APIs to support product discovery and personalized recommendations.
- Outcome: an end-to-end shopping experience combining a branded storefront, interactive product browsing and AI-assisted customer support.

Real-Time Object Detection Dashboard

YOLOv8 | OpenCV | Streamlit | WebRTC | ESP32

- Built a live object-detection application with a Streamlit dashboard and WebRTC webcam support; extended the feed with an ESP32-connected camera node.
- Packaged the application as a Windows desktop installer, making the detection interface easier to launch and demonstrate outside the development environment.

AI Support-Ticket Triage & Email Automation

n8n | Gmail API | Slack API | LLM integration

- Built a workflow that classifies incoming support emails, routes messages to topic-specific Slack channels and flags urgency for review.
- Generated contextual draft responses to streamline repetitive correspondence while allowing a person to review the suggested reply.

EEG Data Visualization Application - Prototype Design

ESP32 data interface | Python backend | SQL | Browser dashboard

- Defined a full stack prototype for receiving EEG sample streams, storing timestamped sessions and presenting waveform views through a browser interface.
- Project scope covers frontend/backend integration, session history and data-quality views for engineering research and signal visualization.

TRAINING & PROFESSIONAL DEVELOPMENT

atomcamp AI Bootcamp

Practical training in AI, generative models and intelligent automation

atomcamp provides practical AI training that connects technical learning with real-world application development.

My atomcamp AI Bootcamp journey developed my ability to turn AI concepts into practical applications through hands-on learning, projects and automation workflows.

- **AI foundations:** deep learning, NLP, LLMs and model development, with practical work in Python and modern machine-learning libraries.
- **Generative & Agentic AI:** RAG, LangChain, LangGraph, CrewAI, API-connected tools and workflow automation with n8n.
- **Application delivery:** FastAPI, Docker, CI/CD and cloud deployment fundamentals for serving and maintaining AI-powered applications.

ADDITIONAL TRAINING & CERTIFICATIONS

CS50x - Harvard University | Computer science, algorithms, data structures and web development.

AI Apps on Azure - Microsoft/edX | Cloud-based AI application development.

Data Analysis with Python - IBM | Data preparation, analysis and model development.

Effective Leadership - Deloitte | Communication, teamwork and leadership fundamentals.